WEEK 8

Diamond Structure

Final project — two halves, one shape.

최종 프로젝트 — 다이아몬드 구조물 · 영어코딩

BIG PICTURE

Everything you've learned 🏆

Inverted pyramid on the bottom, normal pyramid on top.

거꾸로 피라미드 + 일반 피라미드 = 다이아몬드

HOW TO COUNT

Count the axis — one block, one number

1. Stand on the start block — that spot is `(0, 0, 0)`.
2. **x** → walk **right**. Add **+1** for every block.
3. **y** → jump **up**. Add **+1** for every block.
4. **z** → walk **forward**. Add **+1** for every block.

Say them in order every time: `(x, y, z)` = **across, up, forward**.

한 칸 움직이면 +1 — 오른쪽 x, 위로 y, 앞으로 z. 순서는 늘 (x, y, z)

TYPE IT IN

Put the numbers into Minecraft ⌨️

x	y	z
across →	up ↑	forward ⤴

```
place STONE at (x, y, z)
```

- ✓ Open **Code Builder** (press `C` in the world)
- ✓ Type the numbers in order — `(x, y, z)`
- ✓ Press **Run** ▶ — your block appears at that spot

코드 빌더(C) 열고 → (x, y, z) 순서로 숫자 넣고 → Run ▶

CONCEPT

Two halves that meet

bottom half

`grow 1 → 9`

top half

`shrink 7 → 1`

They meet at the widest layer (9 × 9) in the middle.

CONCEPT

Why top starts at 7, not 9

The widest layer (9) is built once — by the bottom half.

If the top also started at 9, you'd stack two 9-layers and lose the point.

가장 넓은 층(9)은 아래 반쪽이 이미 만들었으니 위는 7부터

CODE

Bottom half — grow

```
max_size = 9

size = 1
while size <= max_size:
    build_layer(size)
    agent.move(UP, 1)
    size = size + 2   # 1 → 3 → 5 → 7 → 9
```

CODE

Top half — shrink

```
size = max_size - 2   # start at 7, not 9
while size > 0:
    build_layer(size)
    agent.move(UP, 1)
    size = size - 2   # 7 → 5 → 3 → 1
```

One `max_size` variable controls the whole diamond.

PREDICT 🤔

List every layer, bottom to top

Predict first — then click reveal

```
# bottom: grow 1 → 9
# top: shrink 7 → 1
```

Reveal answer 🔍 reveal

VOCAB

New words today

`diamond`	two pyramids joined at the widest layer	다이아몬드
`max_size`	one variable that sets the whole size	최대 크기
`bottom half`	the growing inverted pyramid	아래 반쪽
`top half`	the shrinking normal pyramid	위 반쪽

🐛 DEBUG

What's wrong here?

buggy

```
# bottom grows to 9, top should start at 7
size = 1
while size <= max_size:
    build_layer(size); agent.move(UP, 1); size += 2

size = max_size         # top half
while size > 0:
    build_layer(size); agent.move(UP, 1); size -= 2
```

Think first — then click next.

🐛 DEBUG

The fix

Bug: The top half starts at `max_size` (9), so the 9-layer is built twice and the diamond looks blocky in the middle. Start the top at `max_size - 2`.

fixed

```
size = max_size - 2    # start at 7
while size > 0:
    build_layer(size); agent.move(UP, 1); size -= 2
```

FINAL PROJECT!

Build your diamond 💎

- ✓ Use one `max_size` variable for everything
- ✓ Bottom half grows 1 → 9
- ✓ Top half shrinks 7 → 1
- ✓ Try `max_size = 5` and `11` too
- ✓ Send your final build + walkthrough on KakaoTalk

QUIZ

Why does the top half start at 7, not 9?

- 9 is too big for Minecraft.
- It's a random choice.
- The widest layer (9) is already built by the bottom half.
- 7 builds faster.

QUIZ

QUESTION 2 OF 3

With `max_size = 9`, how tall is the diamond?

- 9 layers (1,3,5,7,9,7,5,3,1).
- 18 layers.
- 5 layers.
- 4 layers.

QUIZ

QUESTION 3 OF 3

Why use one `max_size` variable for both halves?

- Change one number and the whole diamond resizes.
- It makes the agent move up faster.
- It changes the block color.
- Two variables would crash.

QUIZ

Your score

0 / 3

📖 Re-read the slides. You'll get it.

↺ retake

💎

Course complete!

8 weeks: loops, variables, pyramids, a diamond. Next course: Maze Madness.

8주 완성! 다음 코스: 미로 챌린지에서 만나요 🎉